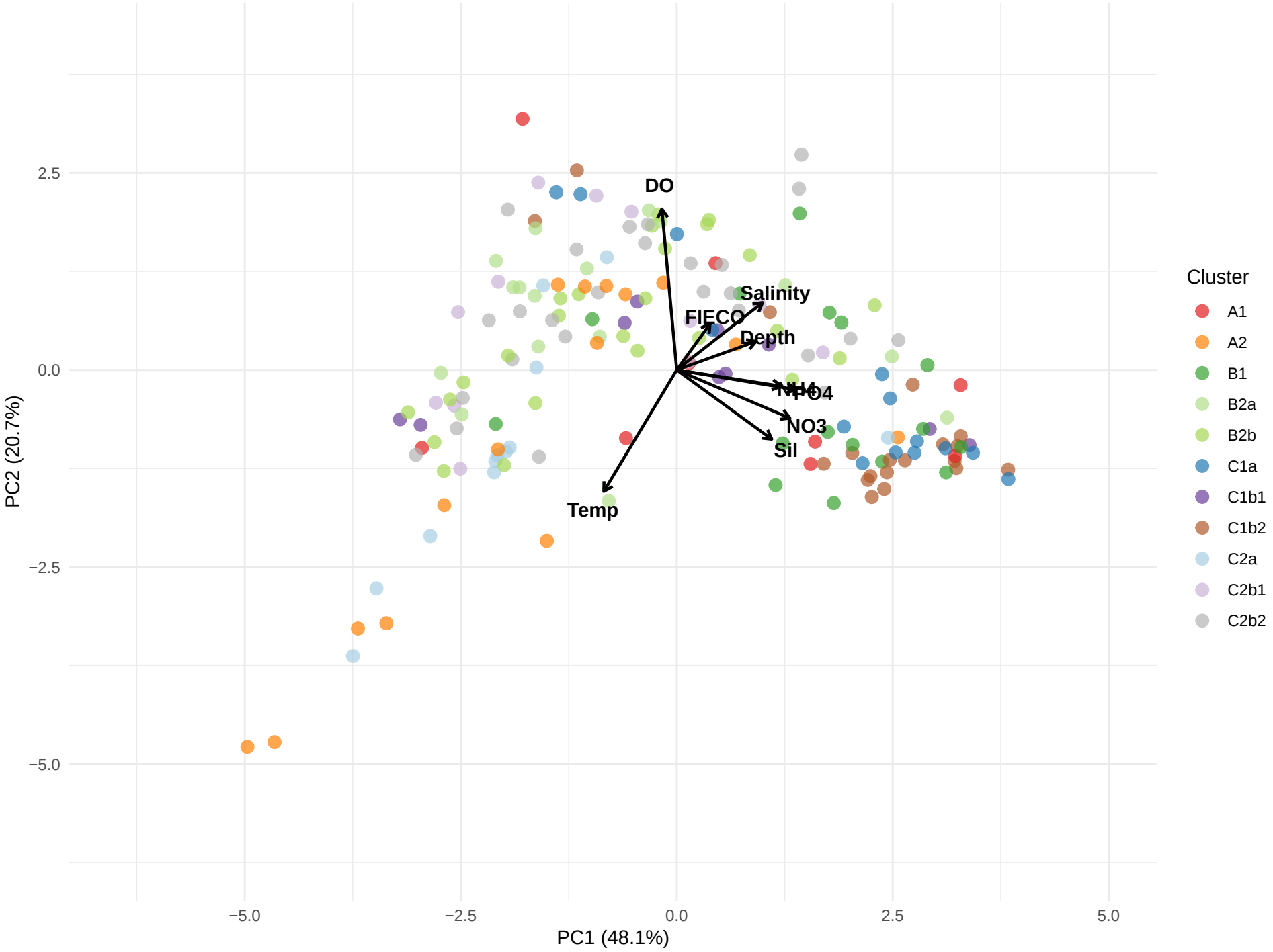
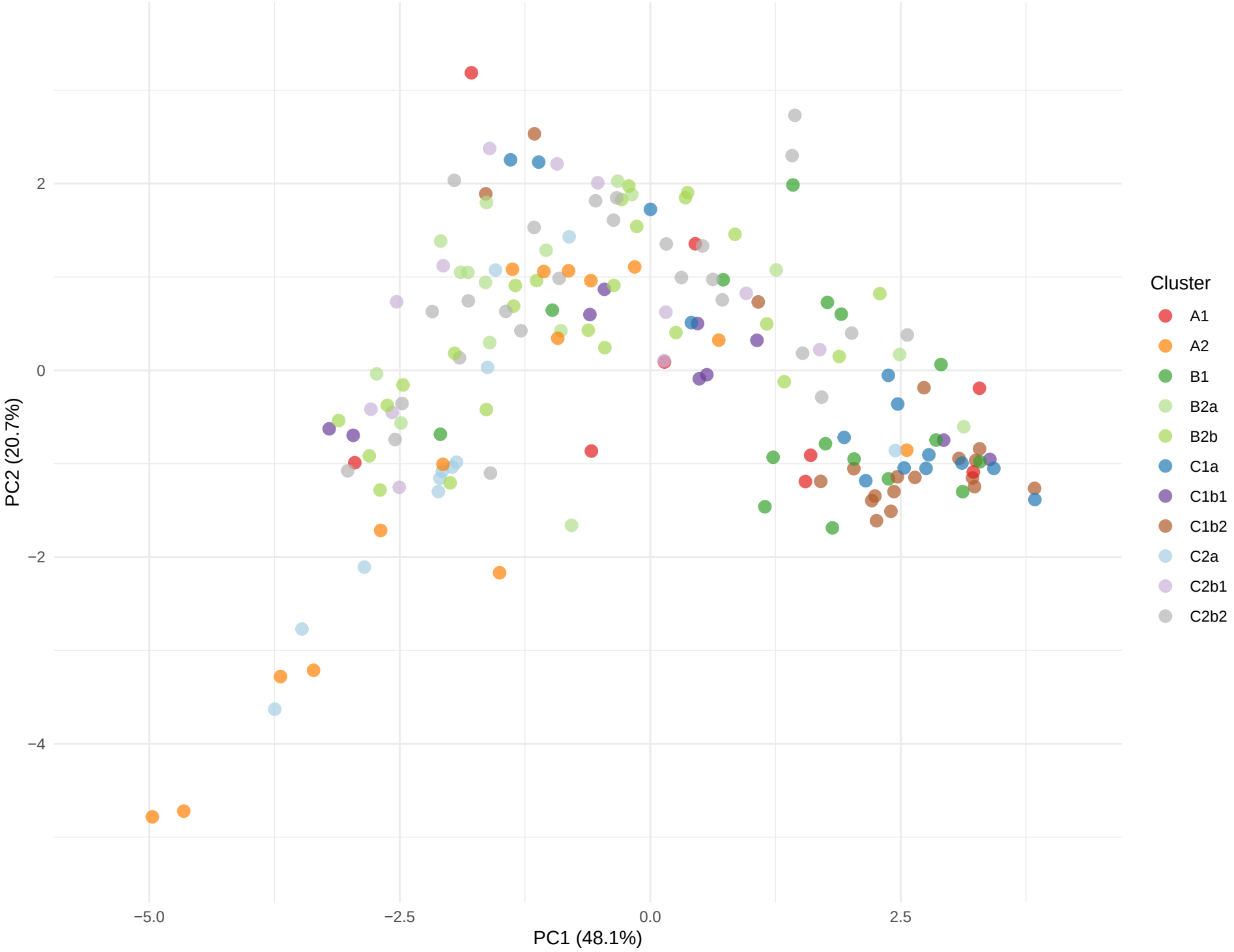
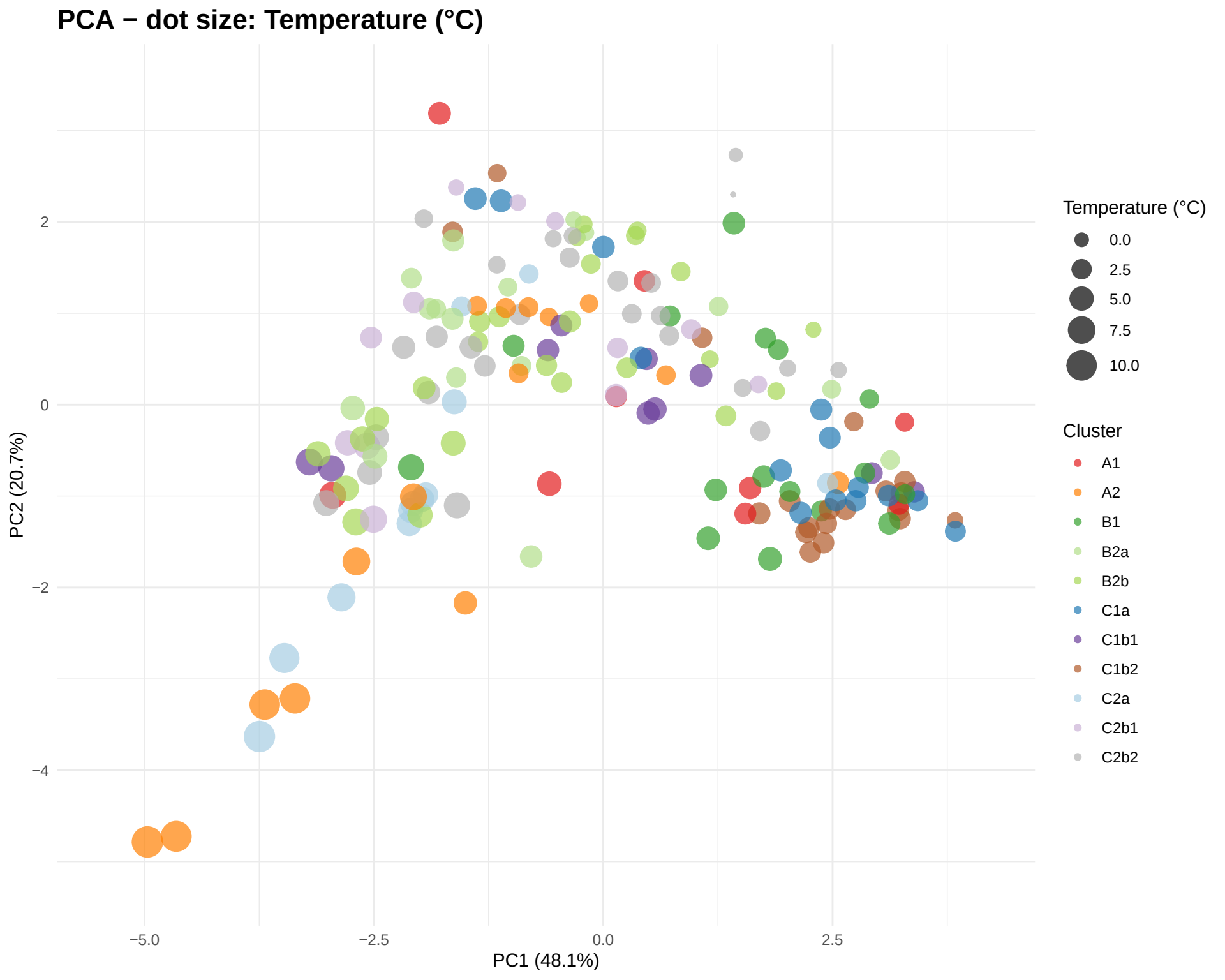
PCA of Environmental Variables (11 clusters)

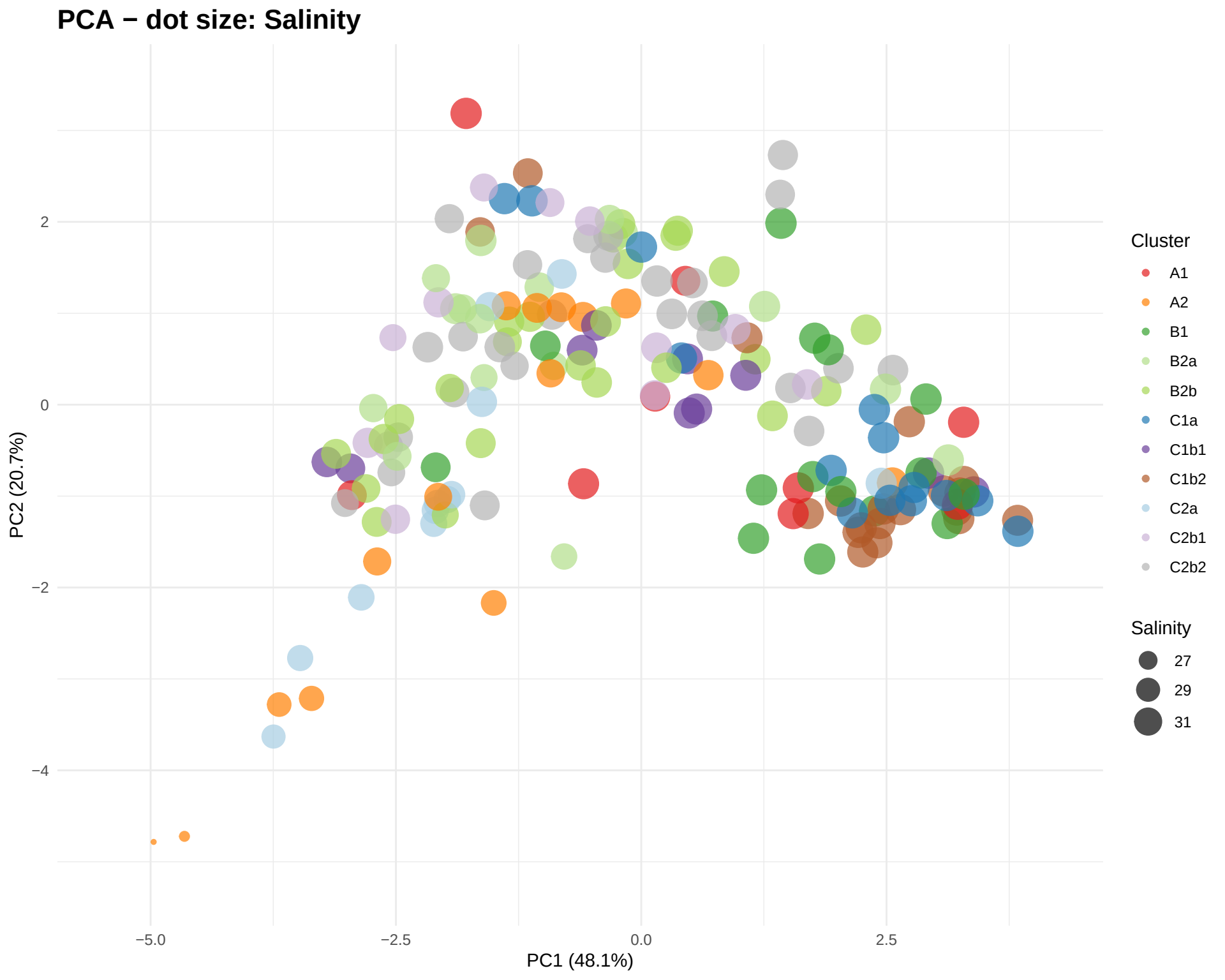
log1p−transformed nutrients + scaled; 9 variables



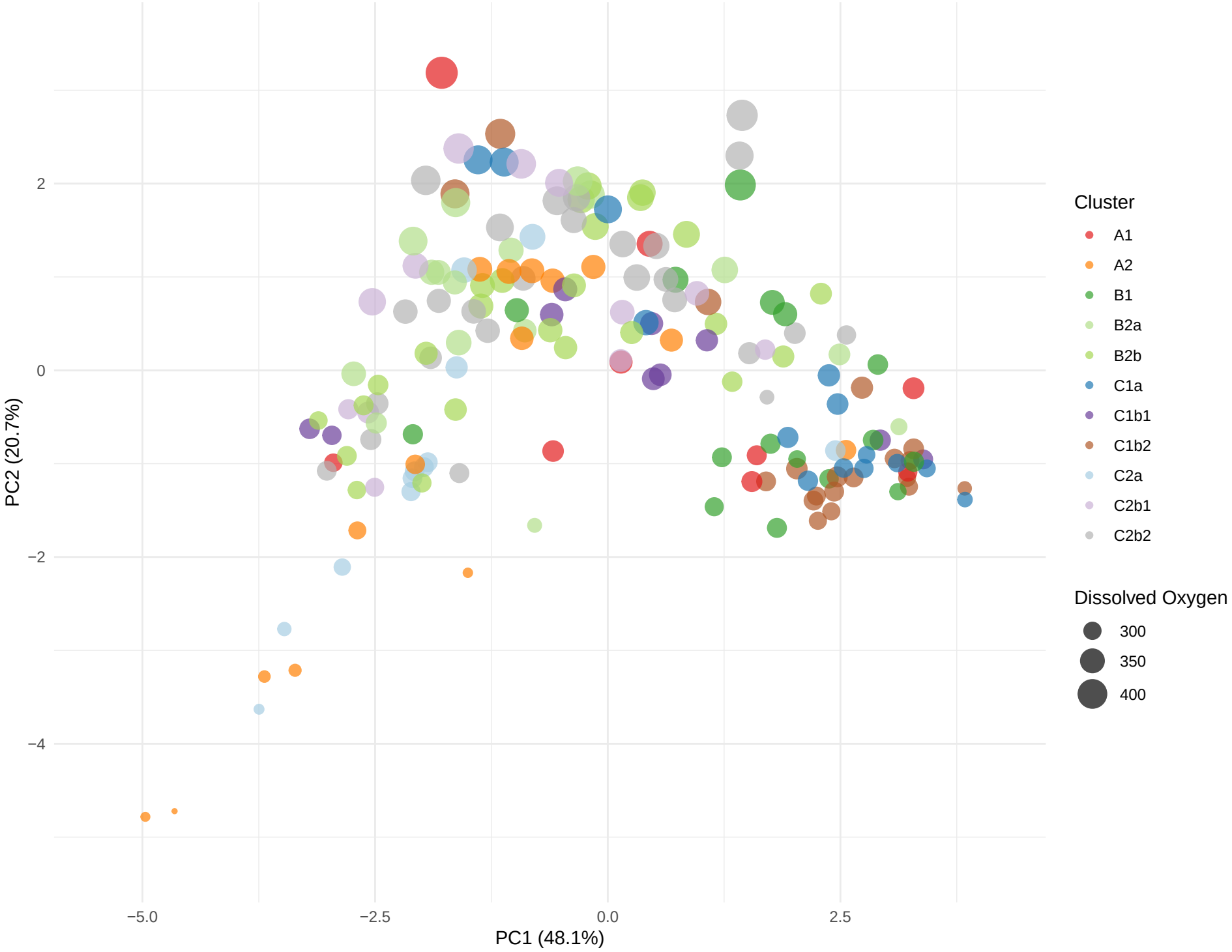
PCA of Environmental Variables (points only)

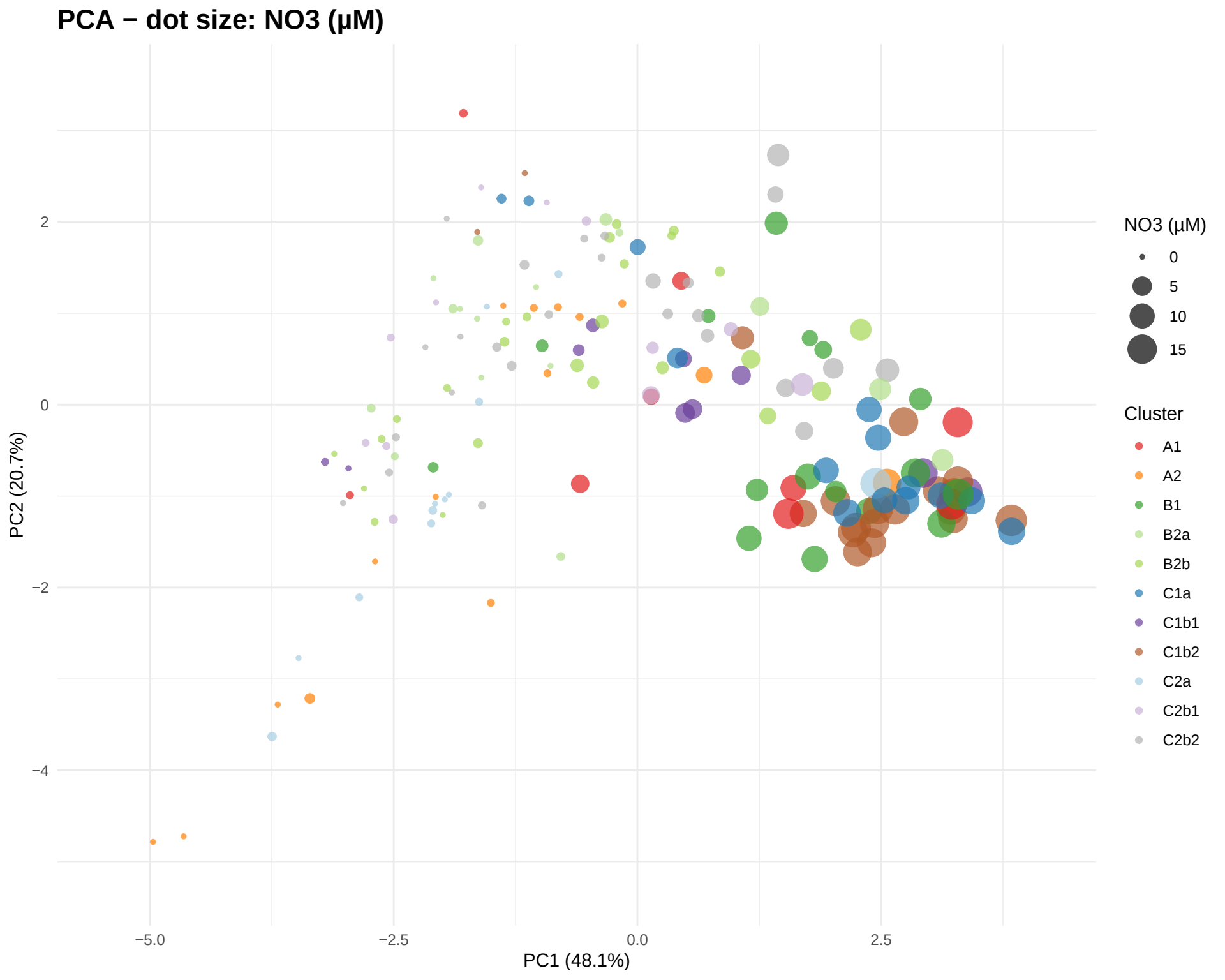


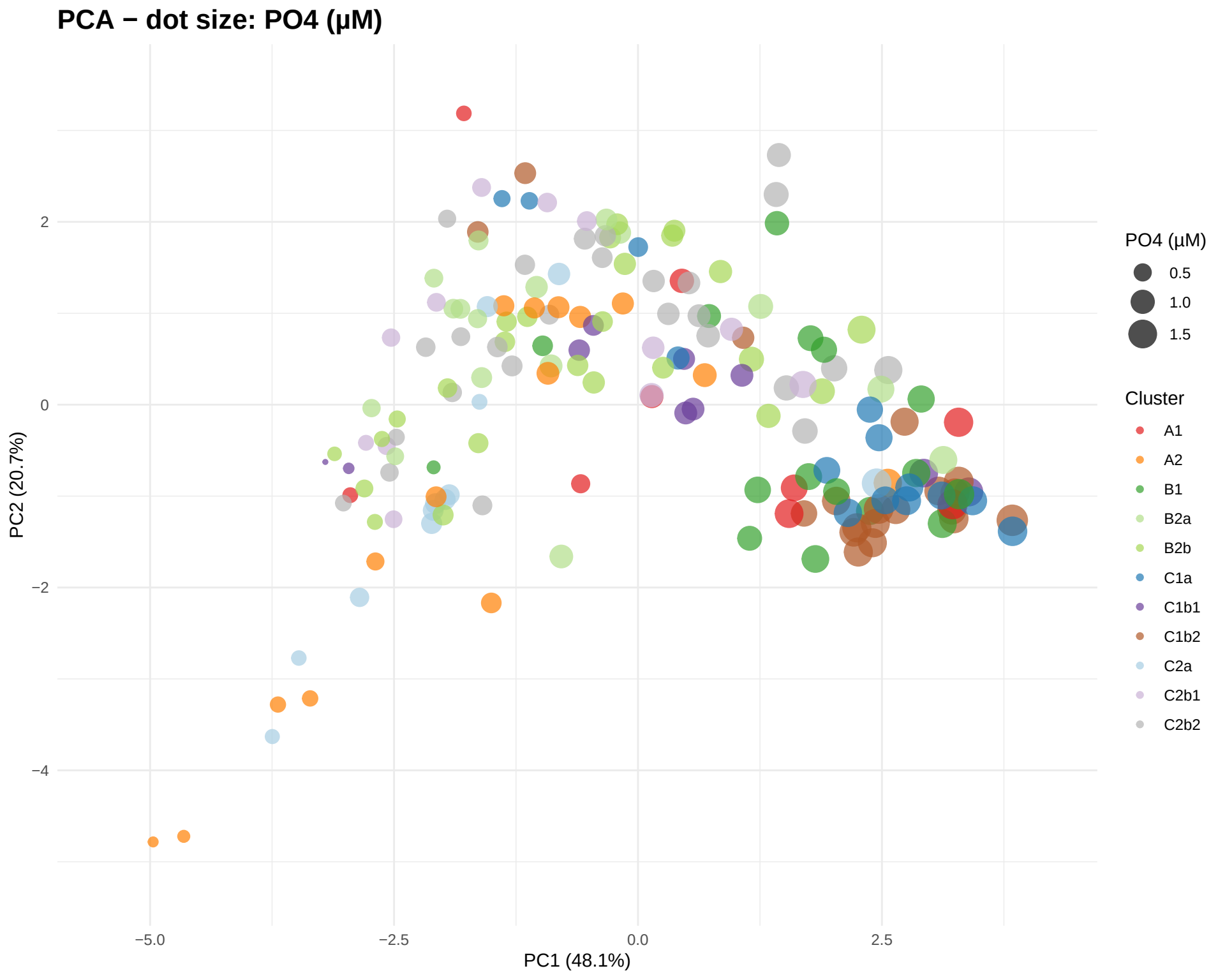

PCA − dot size: Temperature (°C)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Temperature (°C)
0.0
2.5
5.0
7.5
10.0
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2

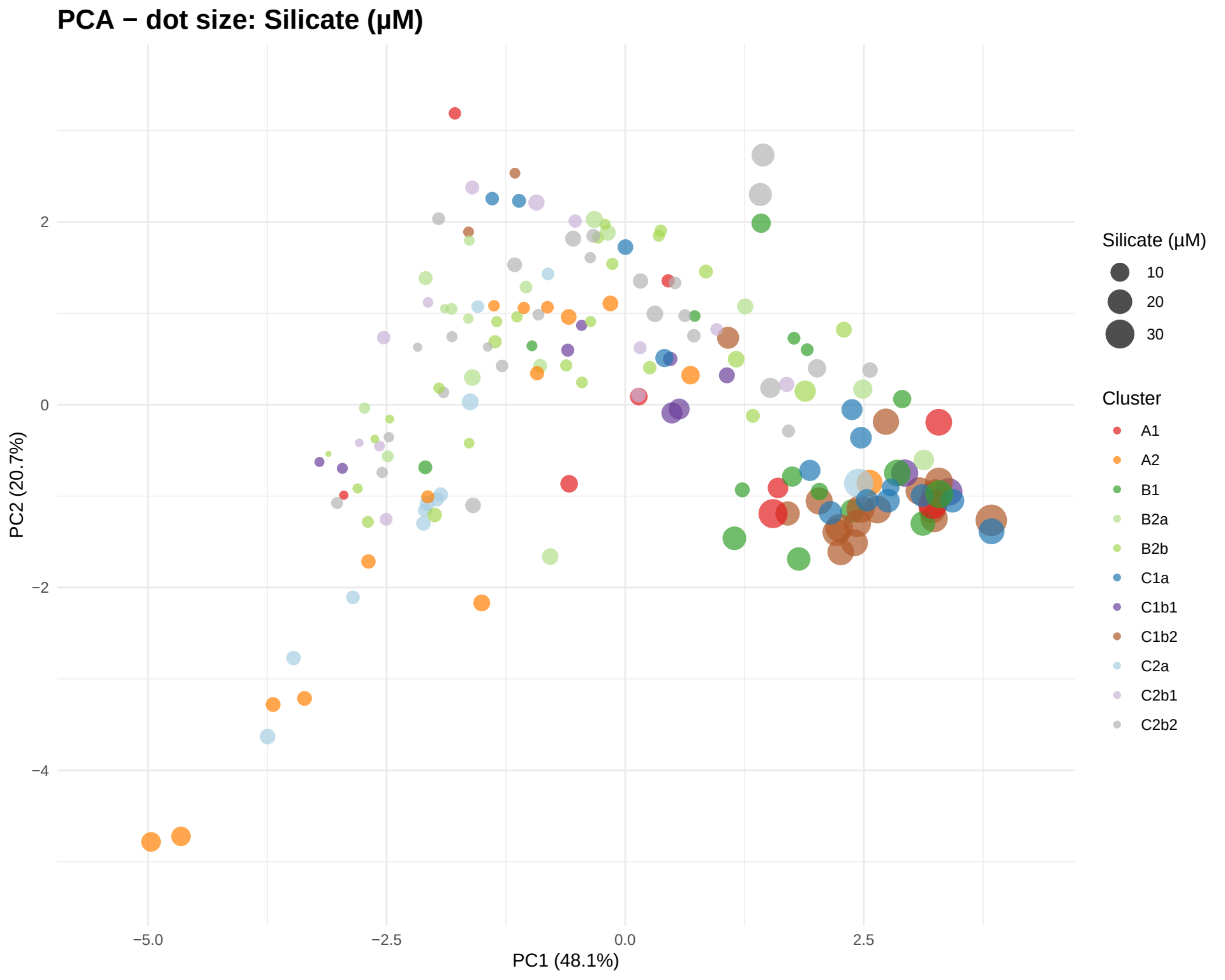

PCA − dot size: Salinity
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
Salinity
27
29
31

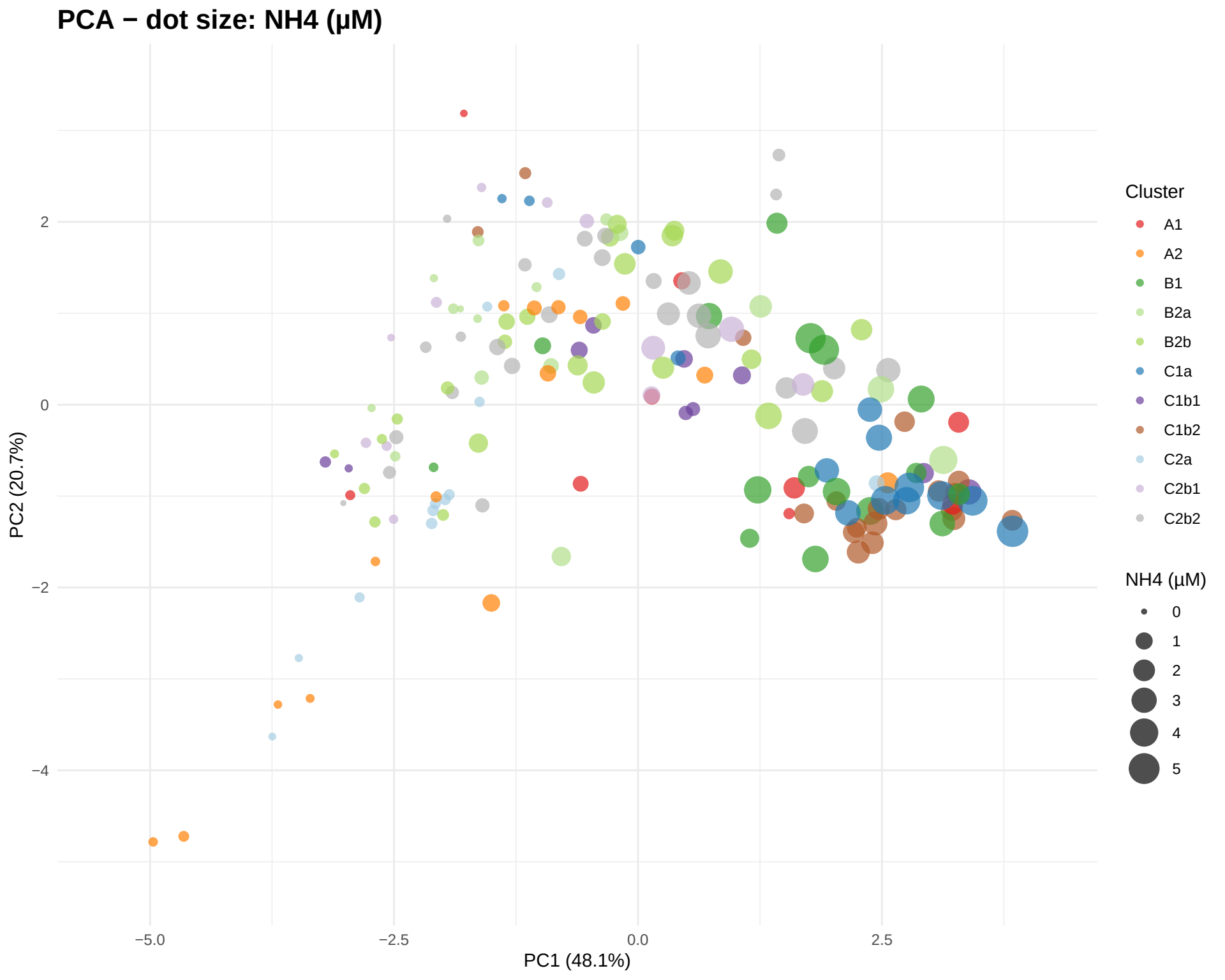
PCA − dot size: Dissolved Oxygen

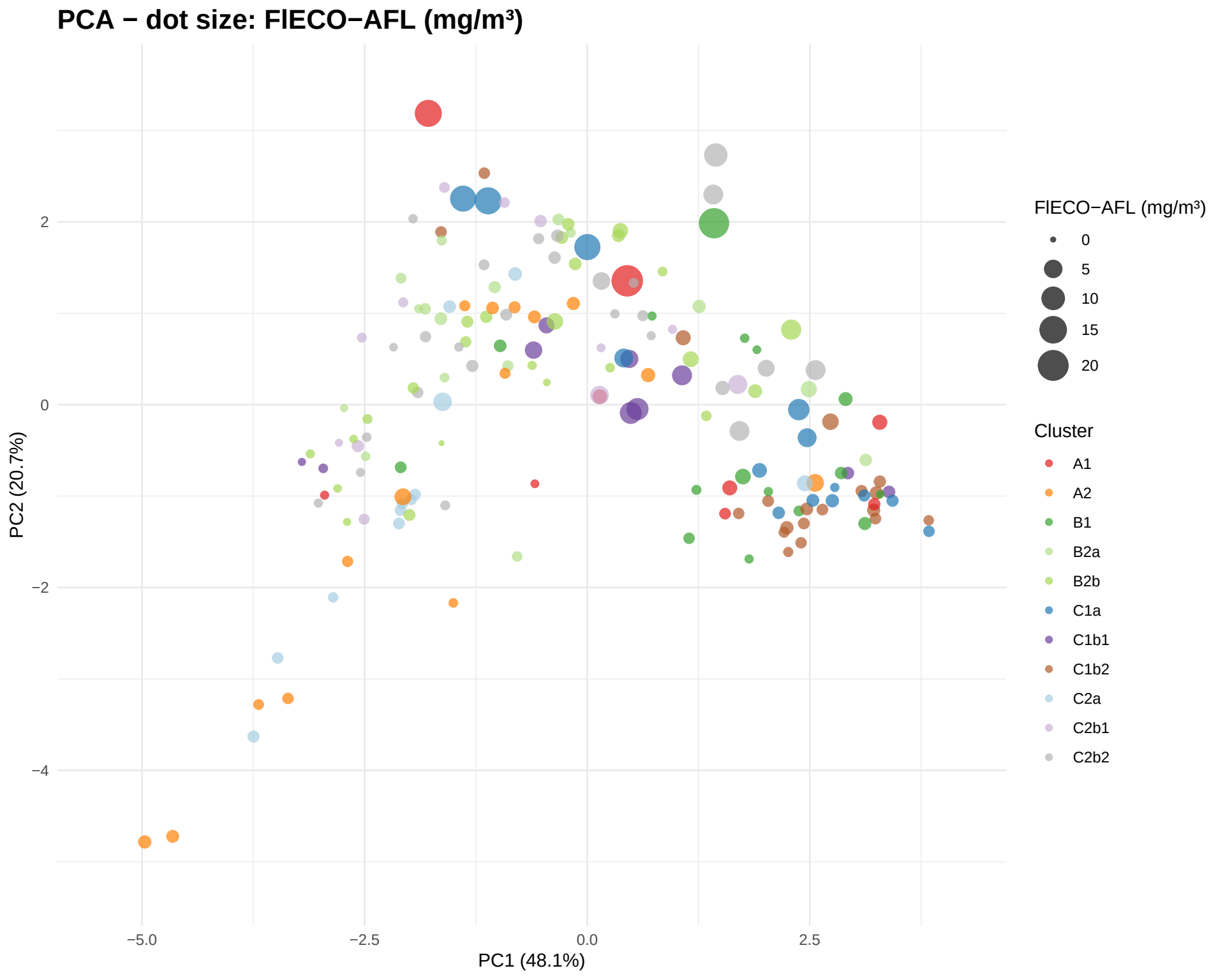


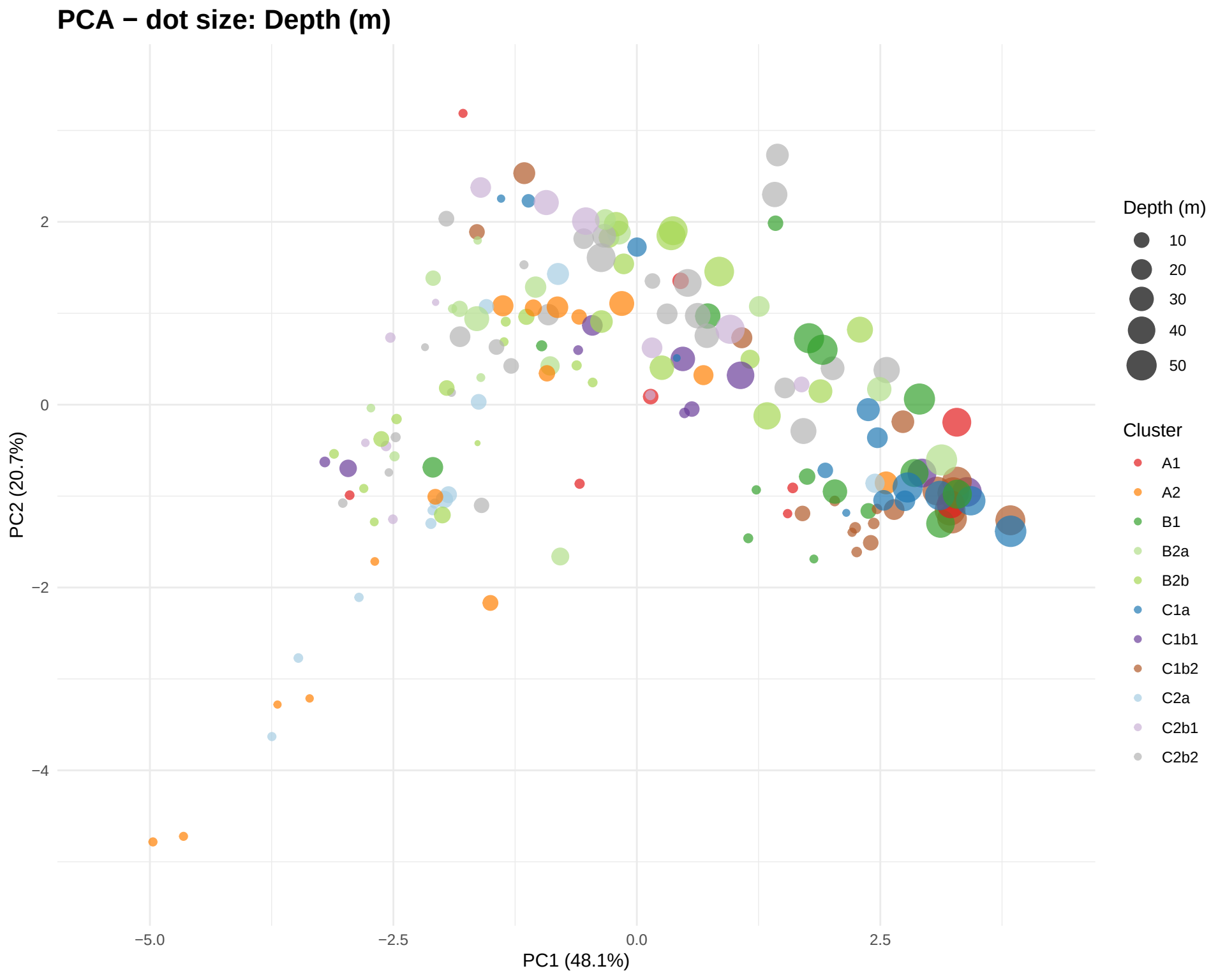

PCA − dot size: NO3 (µM)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
NO3 (µM)
0
5
10
15
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2

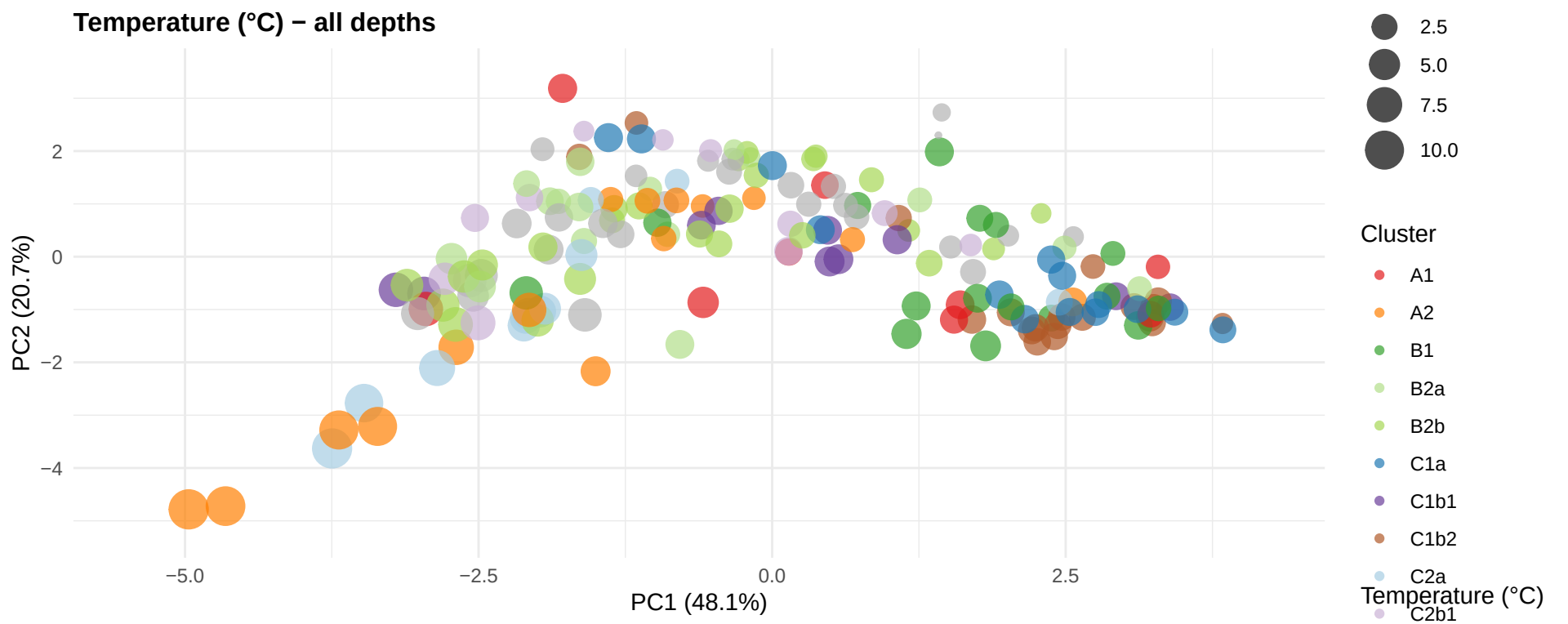

PCA − dot size: PO4 (µM)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
PO4 (µM)
0.5
1.0
1.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2

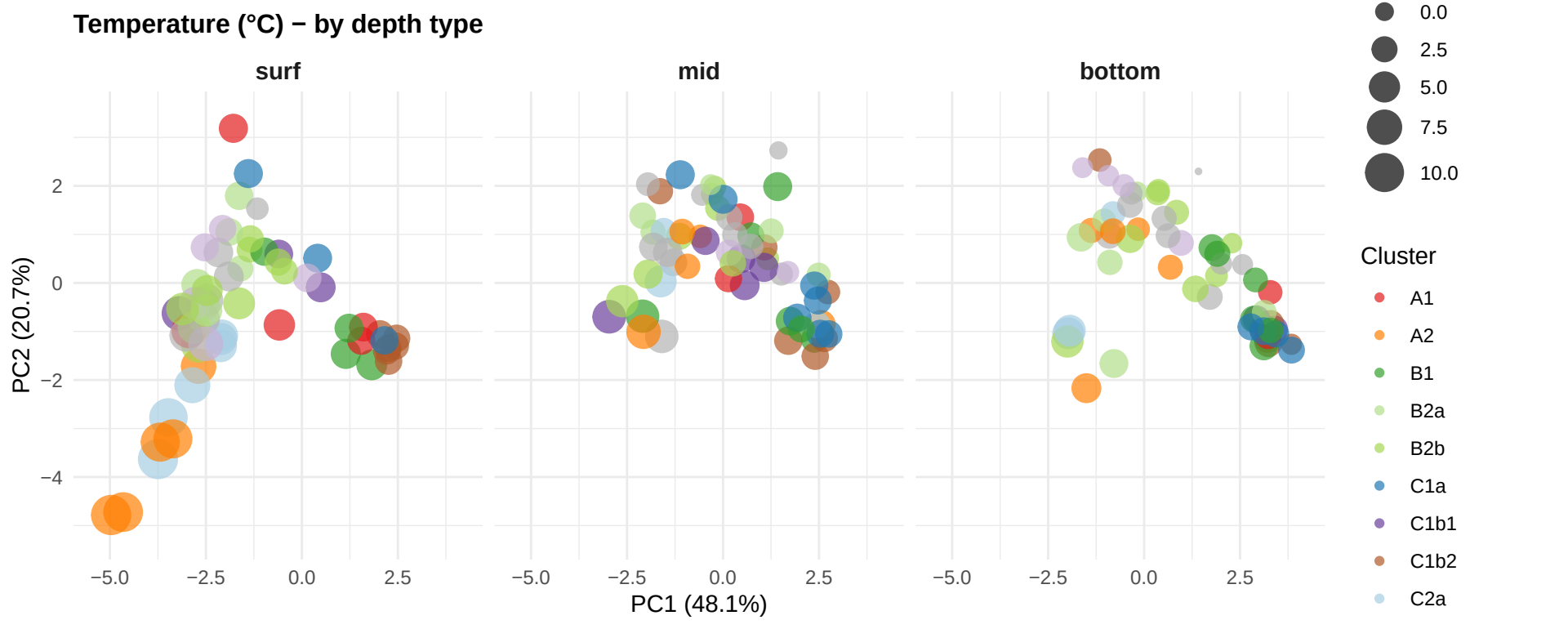

PCA − dot size: Silicate (µM)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Silicate (µM)
10
20
30
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2

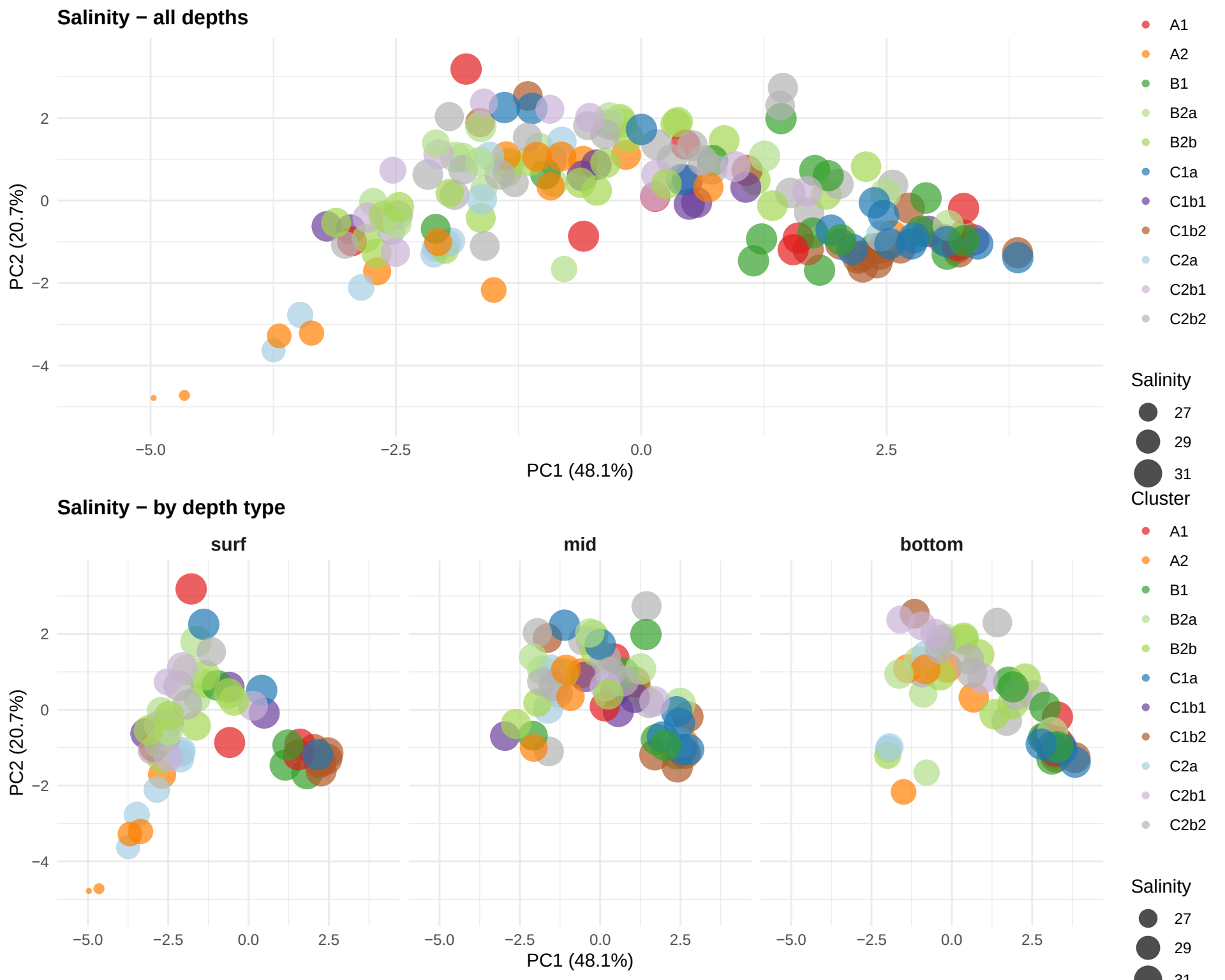

PCA − dot size: NH4 (µM)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
NH4 (µM)
0
1
2
3
4
5

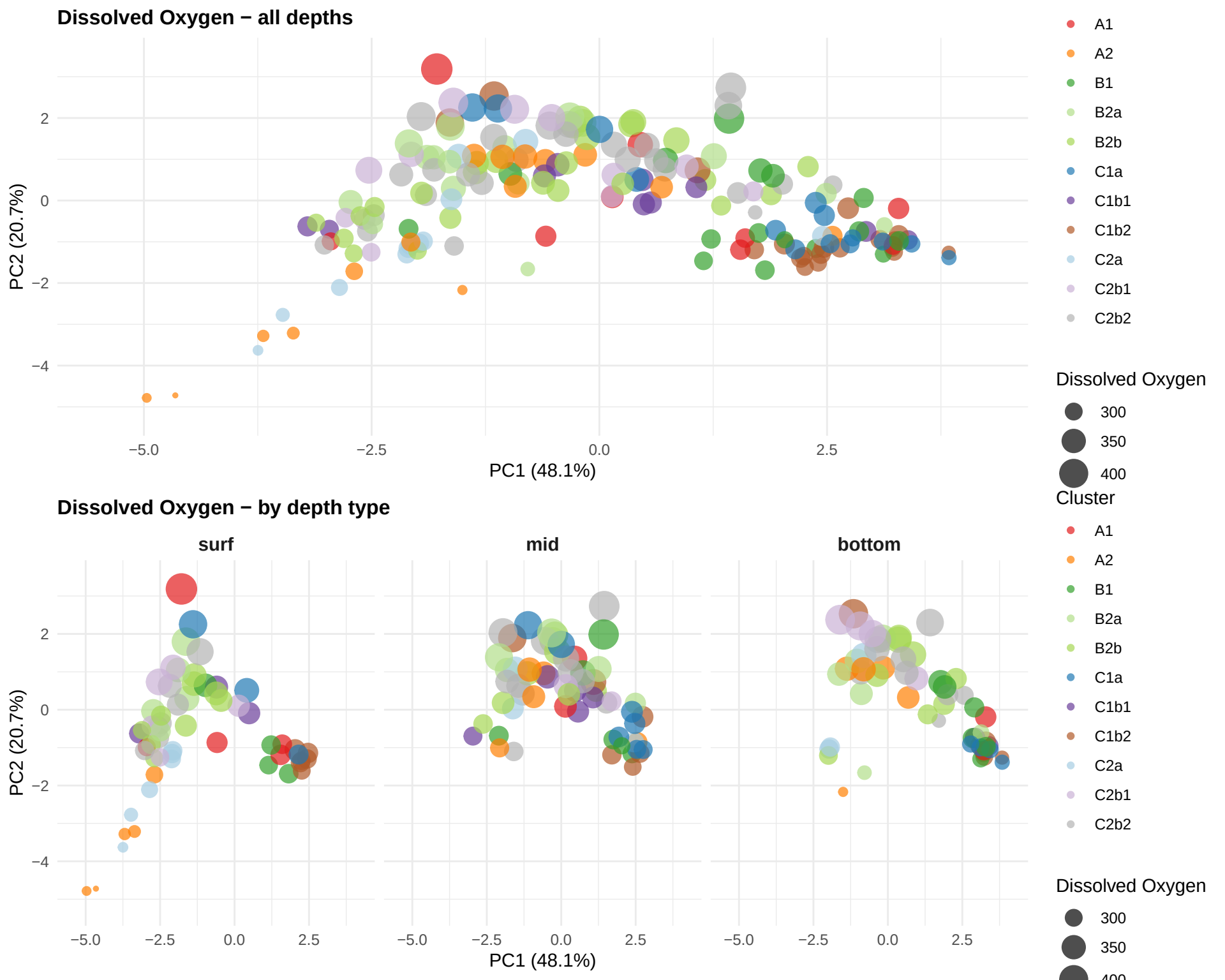

PCA − dot size: FIECO−AFL (mg/m³)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
FIECO−AFL (mg/m³)
0
5
10
15
20
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2

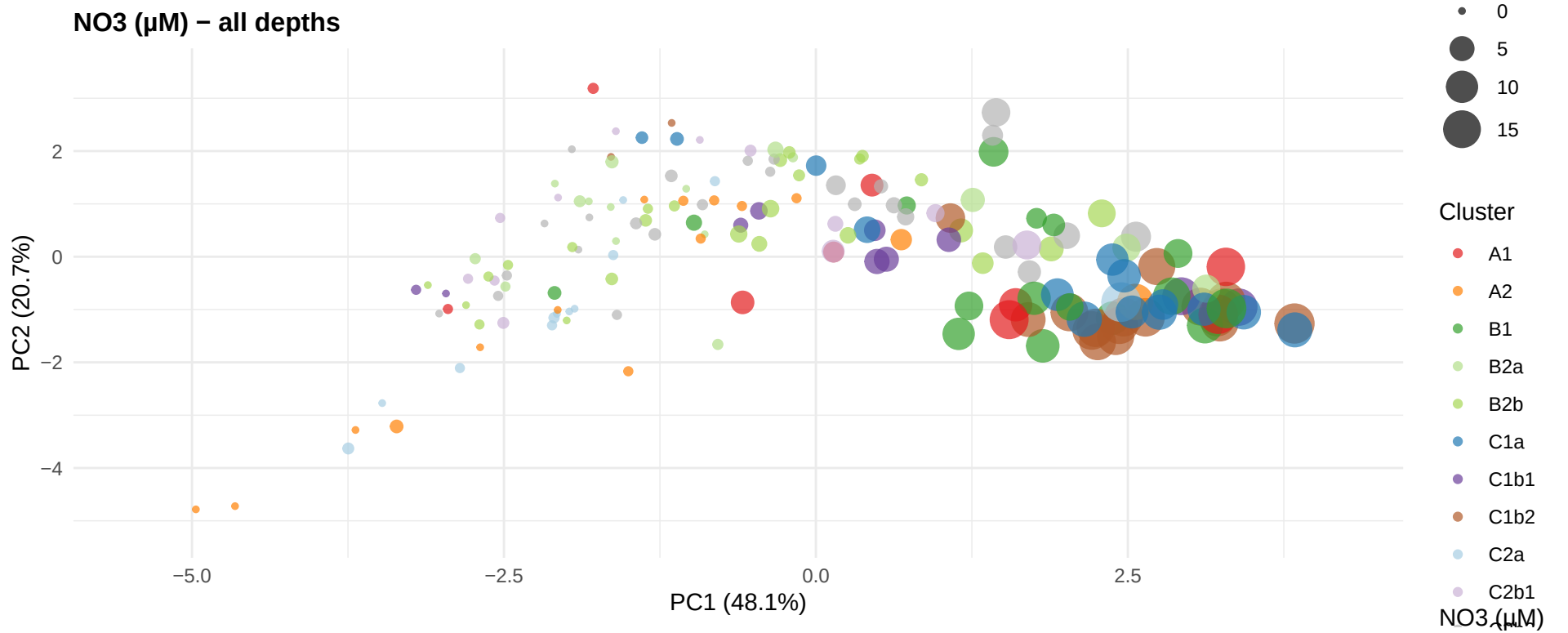

PCA − dot size: Depth (m)
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Depth (m)
10
20
30
40
50
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2

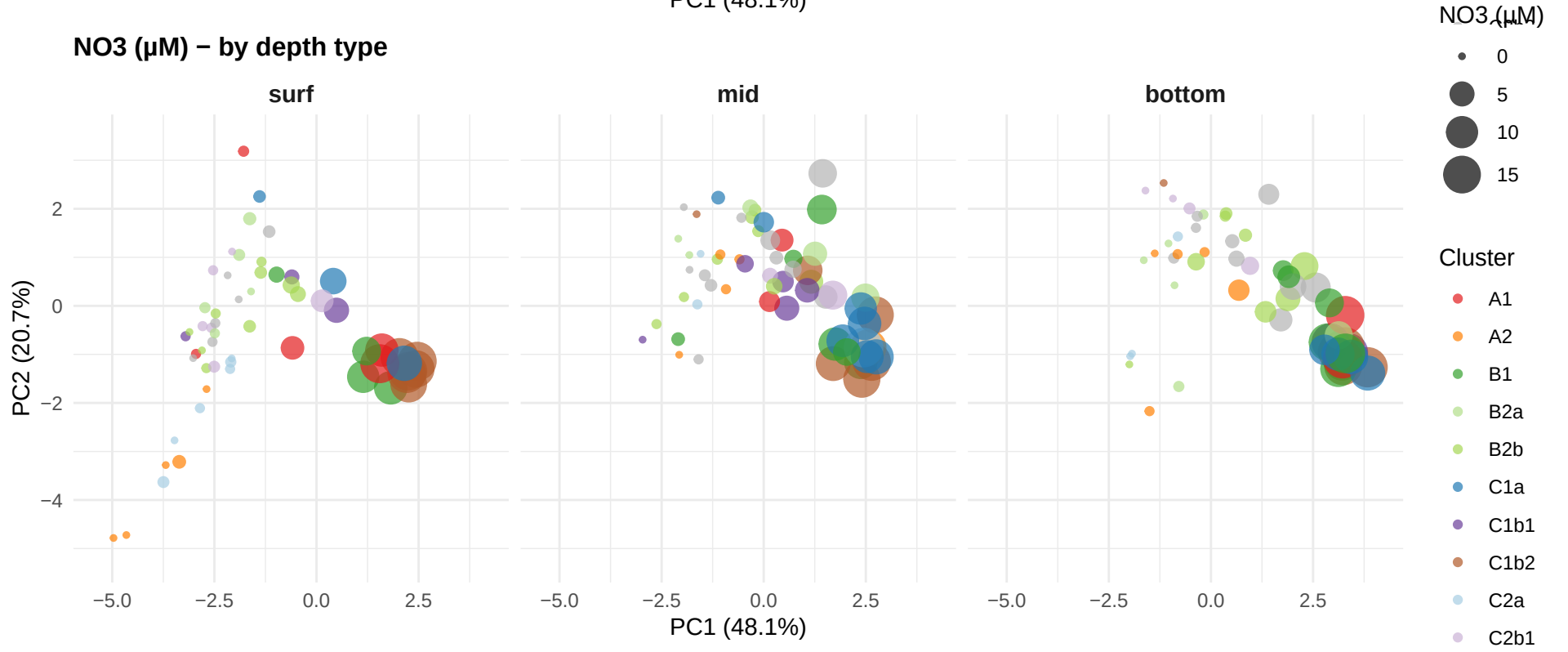

Temperature (°C) − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
2.5
5.0
7.5
10.0
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1

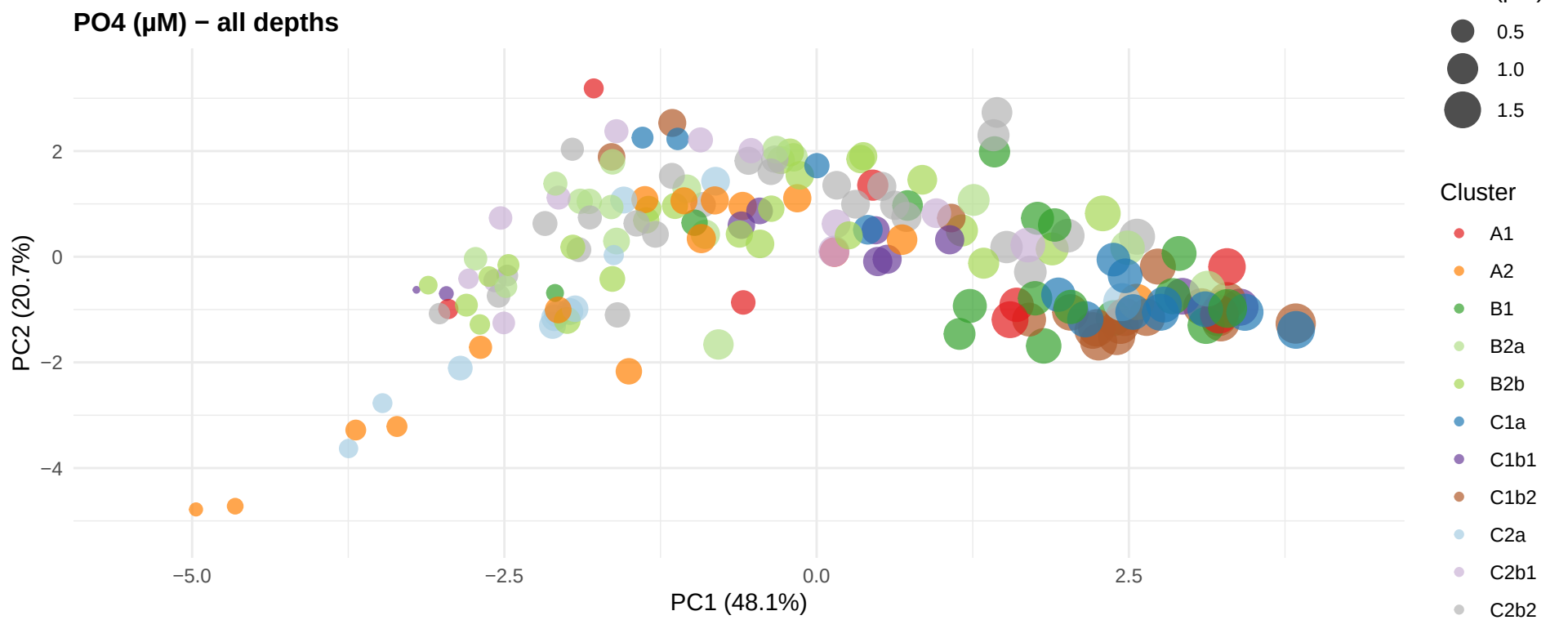

Temperature (°C) − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Temperature (°C)
0.0
2.5
5.0
7.5
10.0
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a

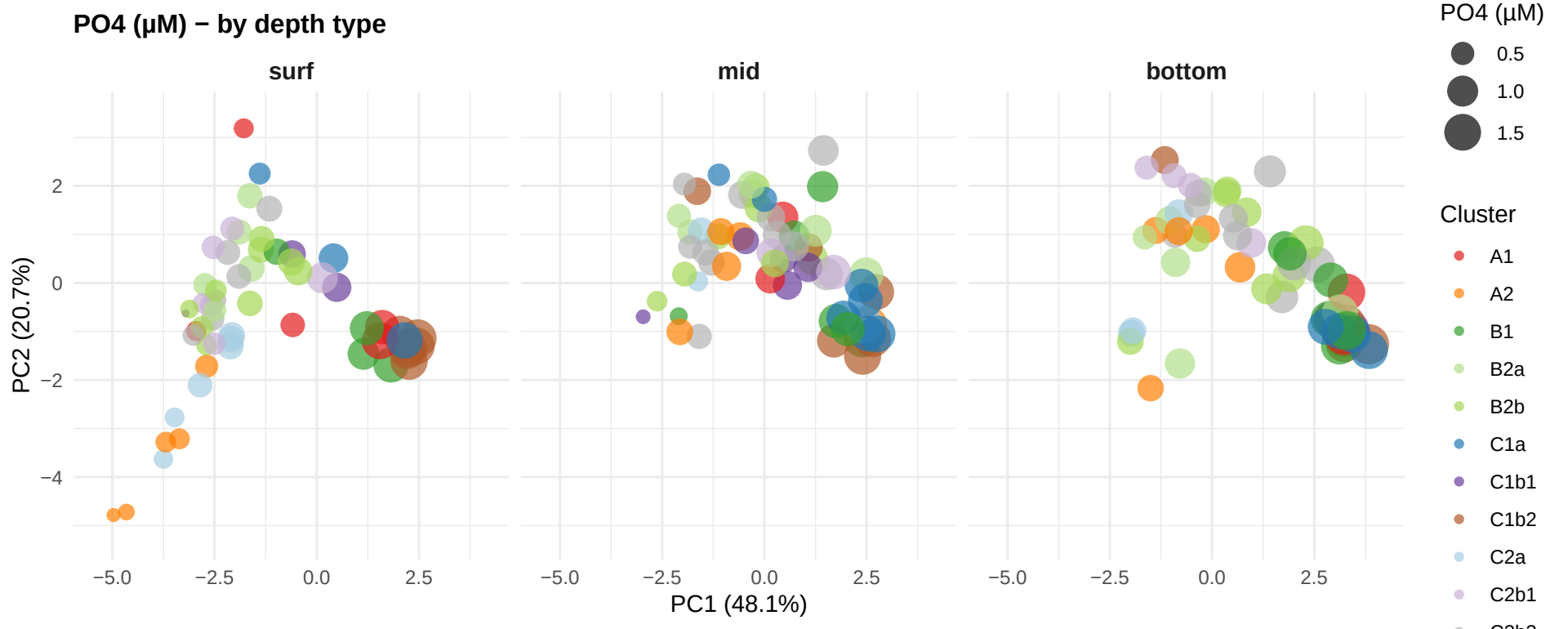

Salinity − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
Salinity
27
29
31
Salinity − by depth type
surf
mid
bottom
Cluster

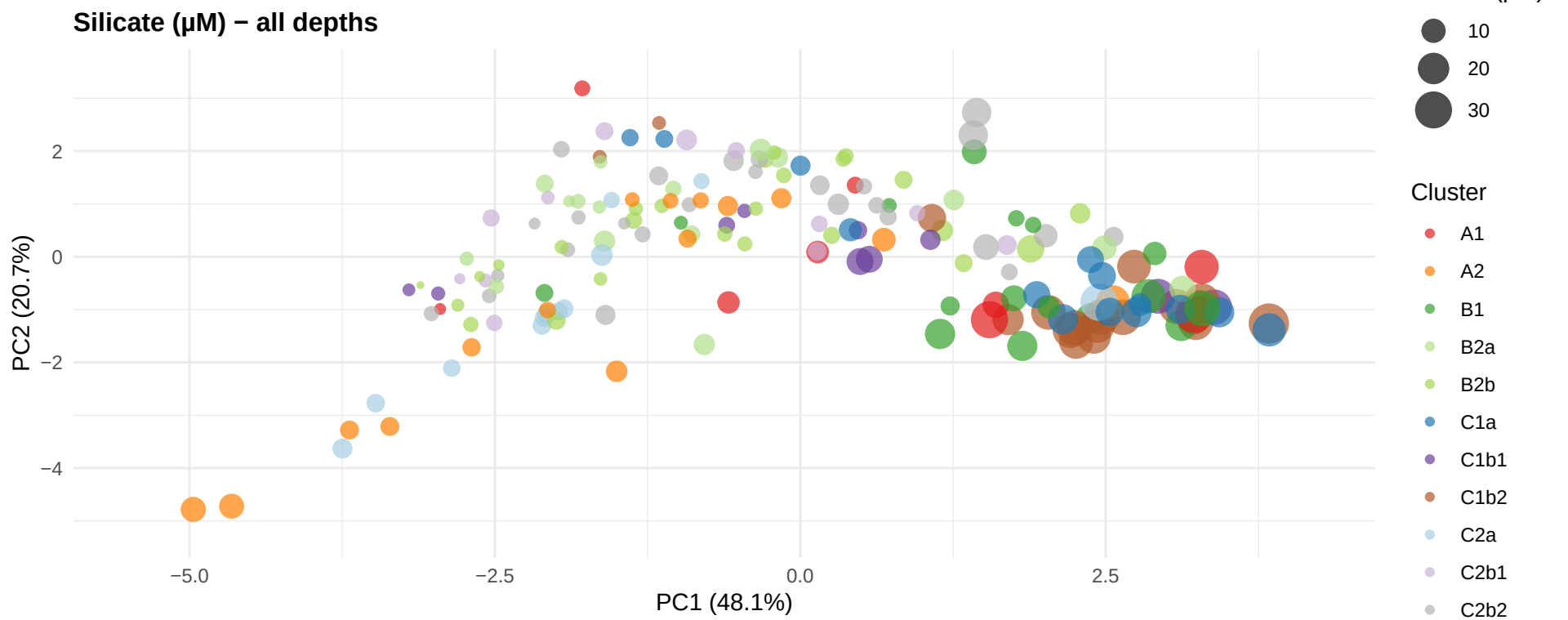

Dissolved Oxygen − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
Dissolved Oxygen
300
350
400
Dissolved Oxygen − by depth type
surf
mid
bottom
Cluster

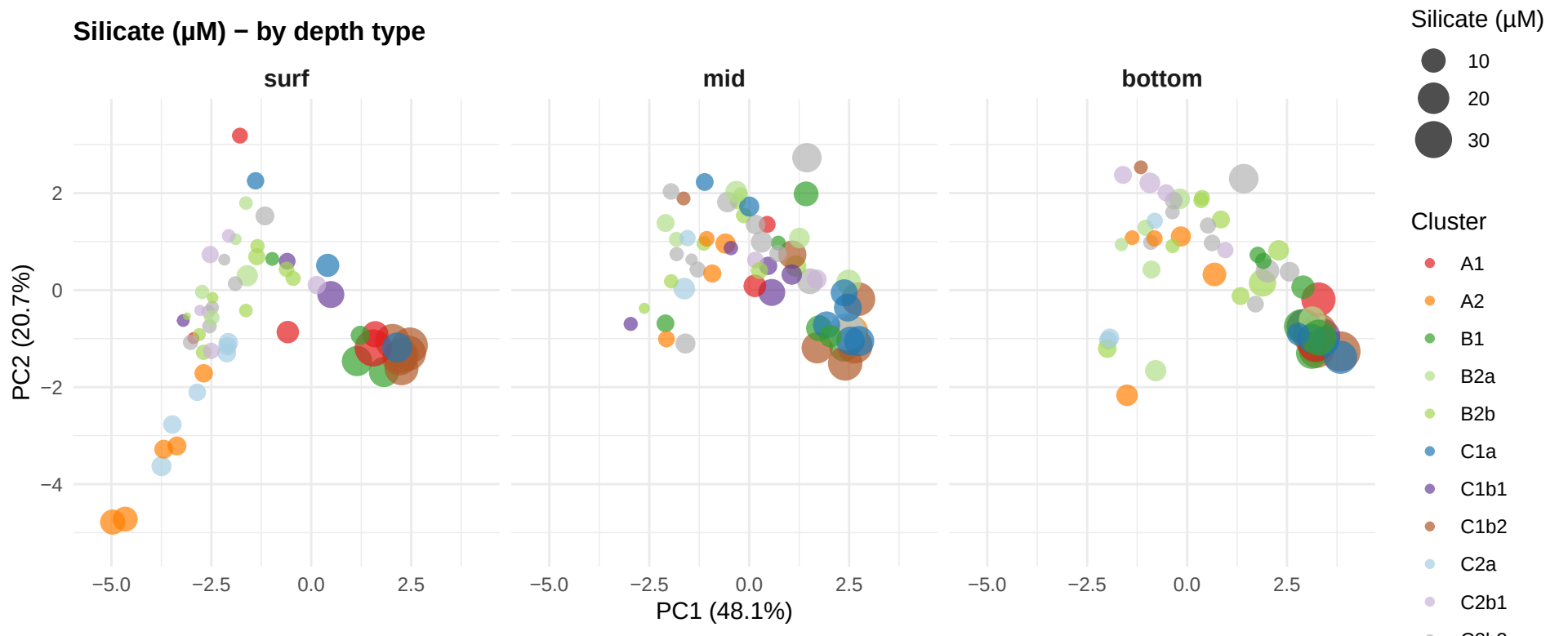

NO3 (µM) − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
0
5
10
15
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1

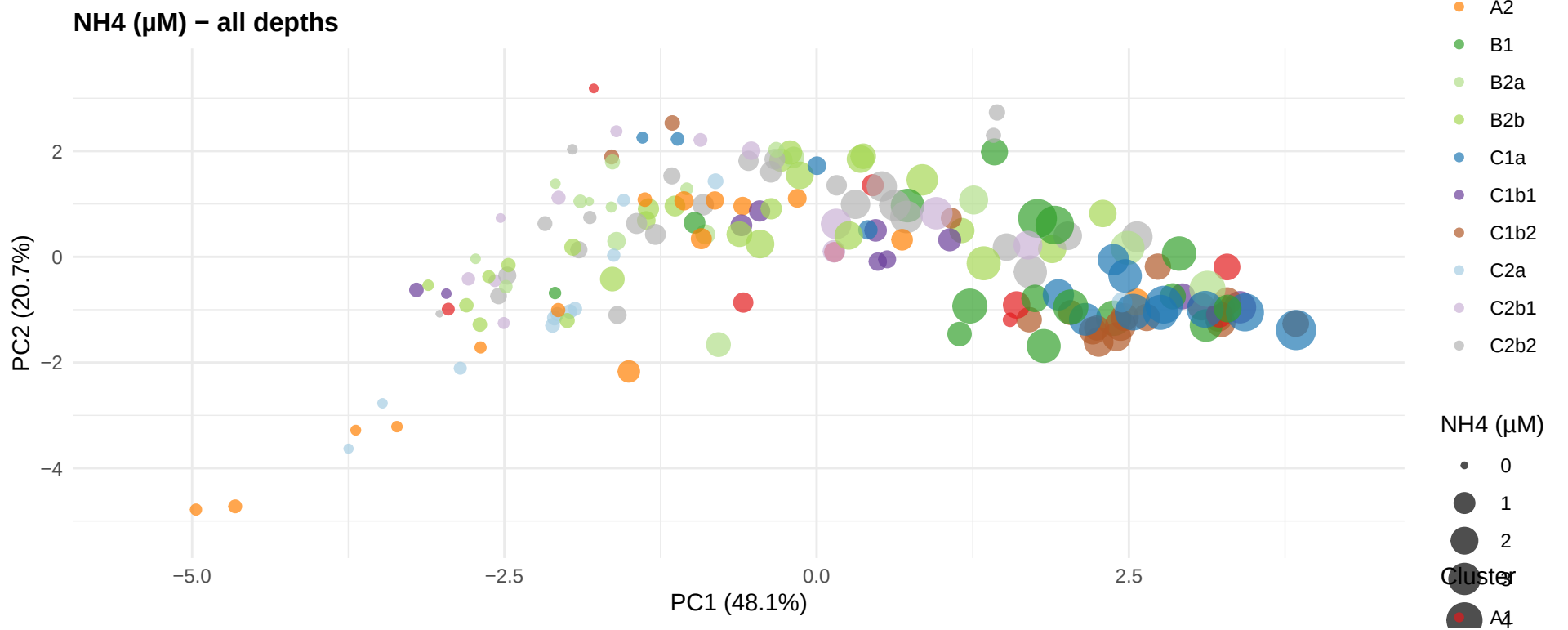

NO3 (µM) − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
NO3 (µM)
0
5
10
15
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1

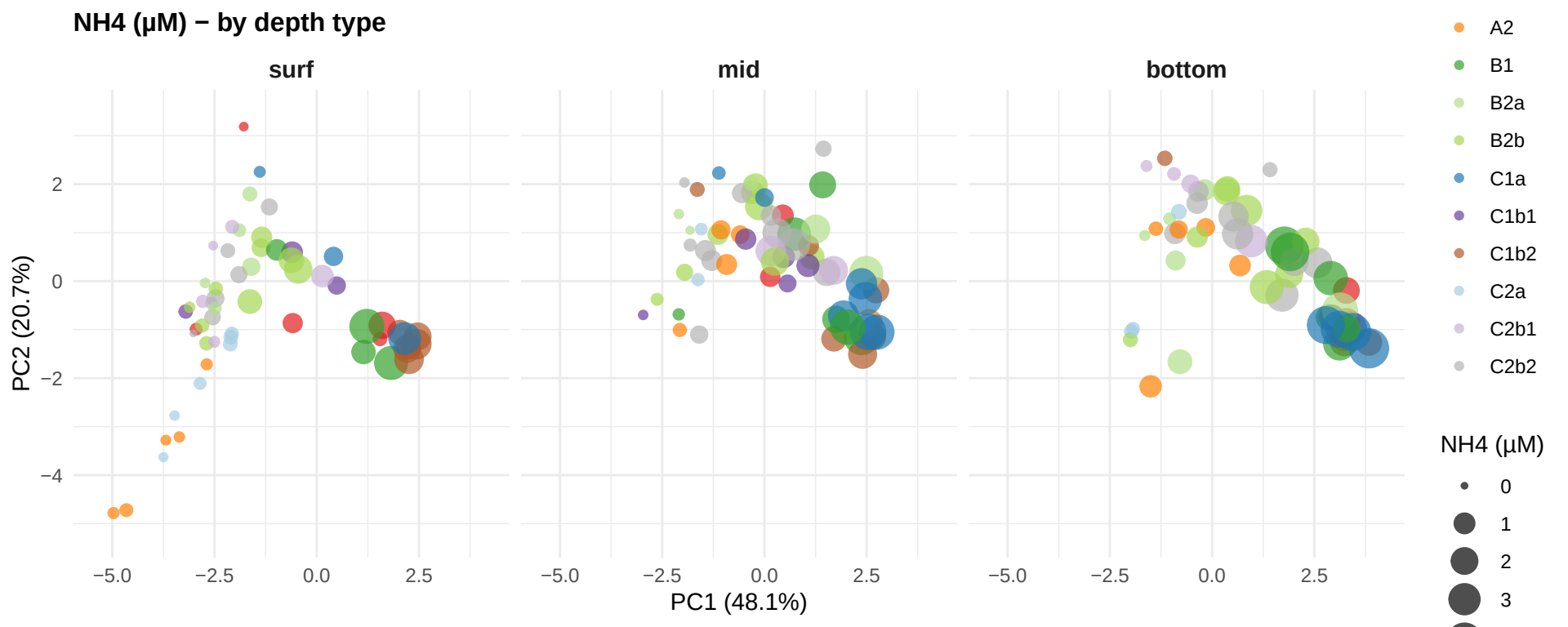

PO4 (µM) − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
0.5
1.0
1.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2

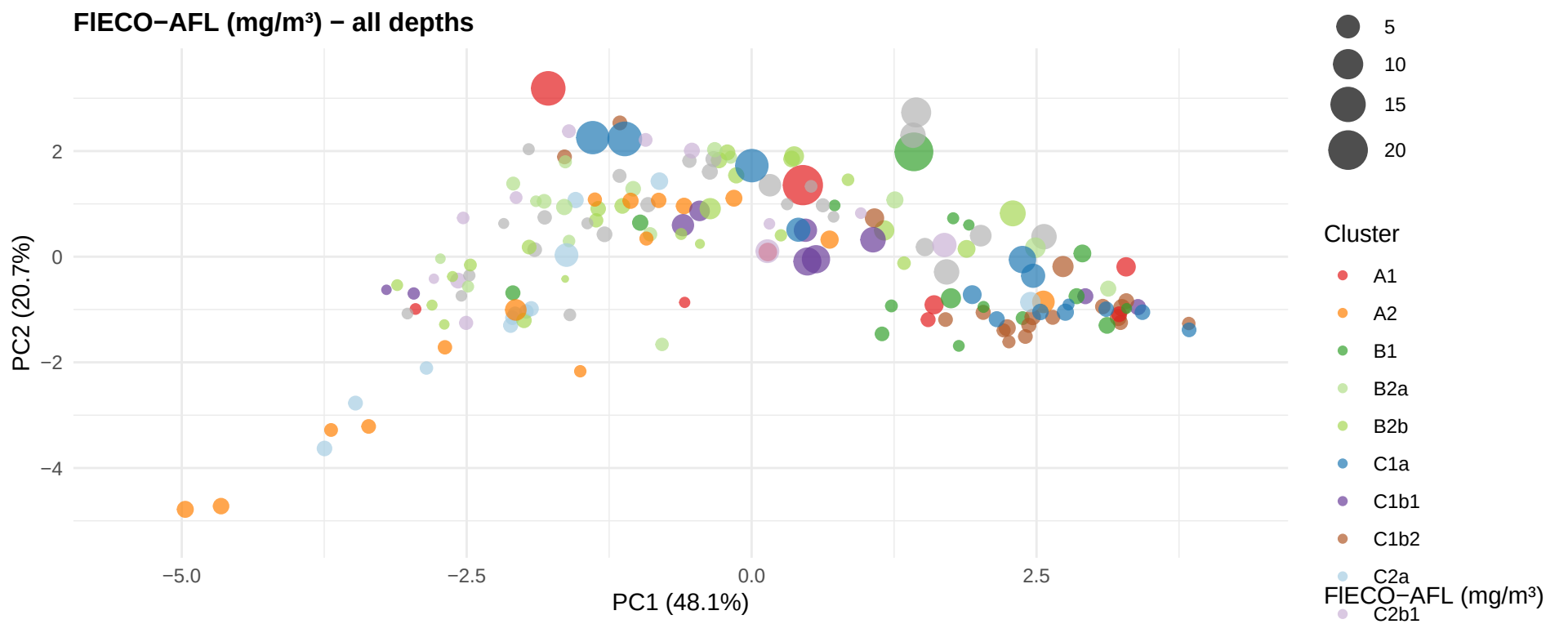

PO4 (µM) − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
PO4 (µM)
0.5
1.0
1.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1

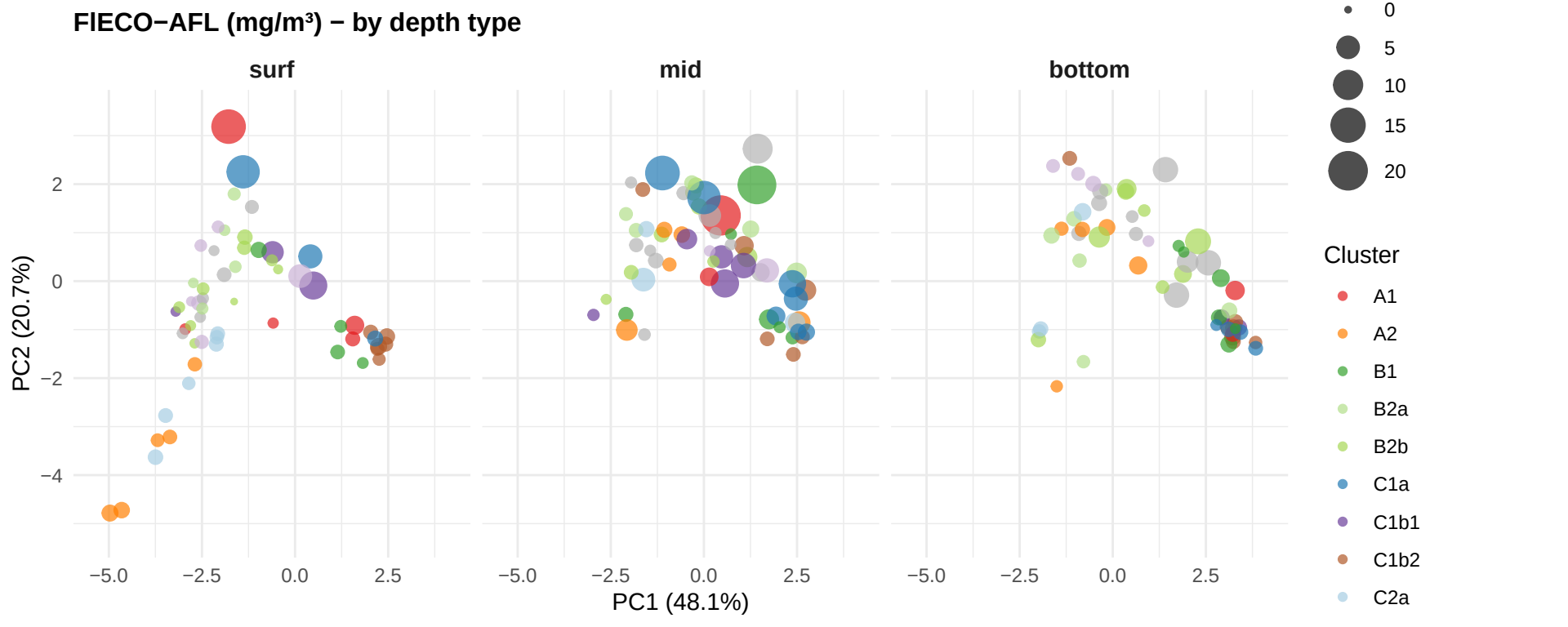

Silicate (µM) − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
10
20
30
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2

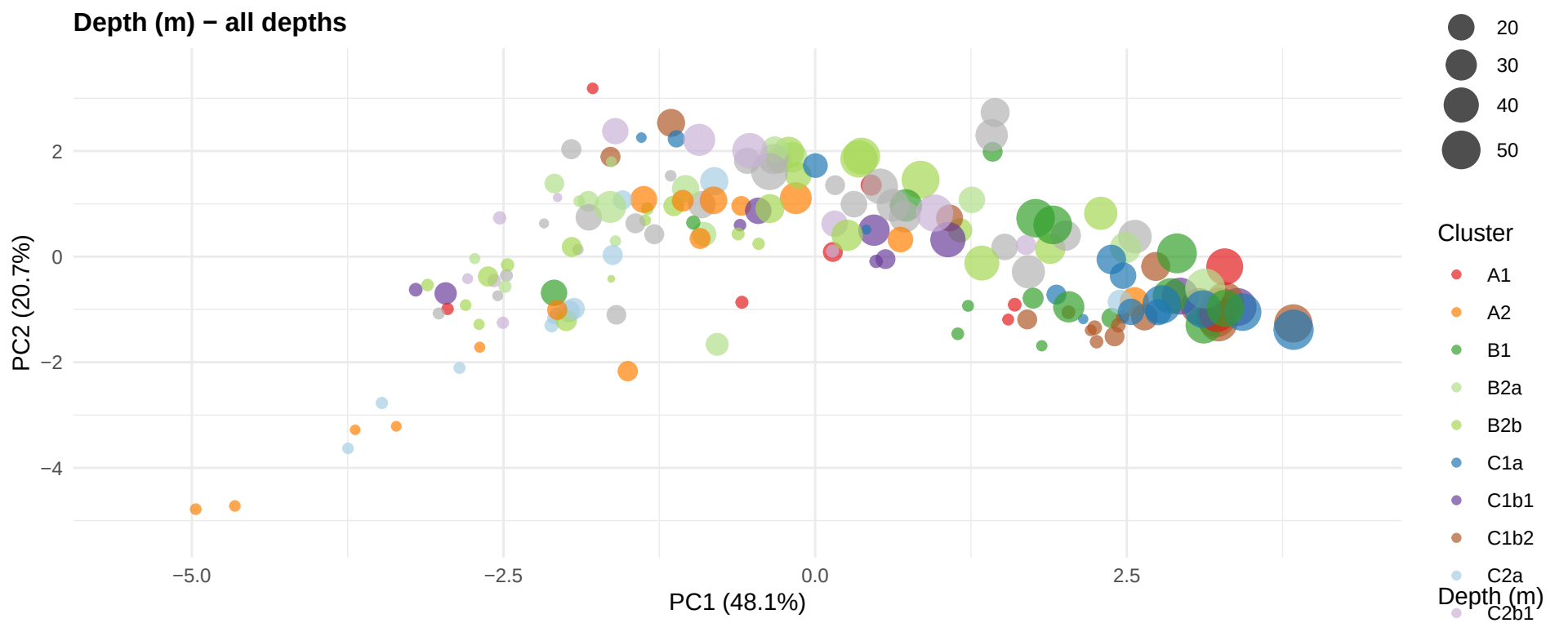

Silicate (µM) − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Silicate (µM)
10
20
30
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1

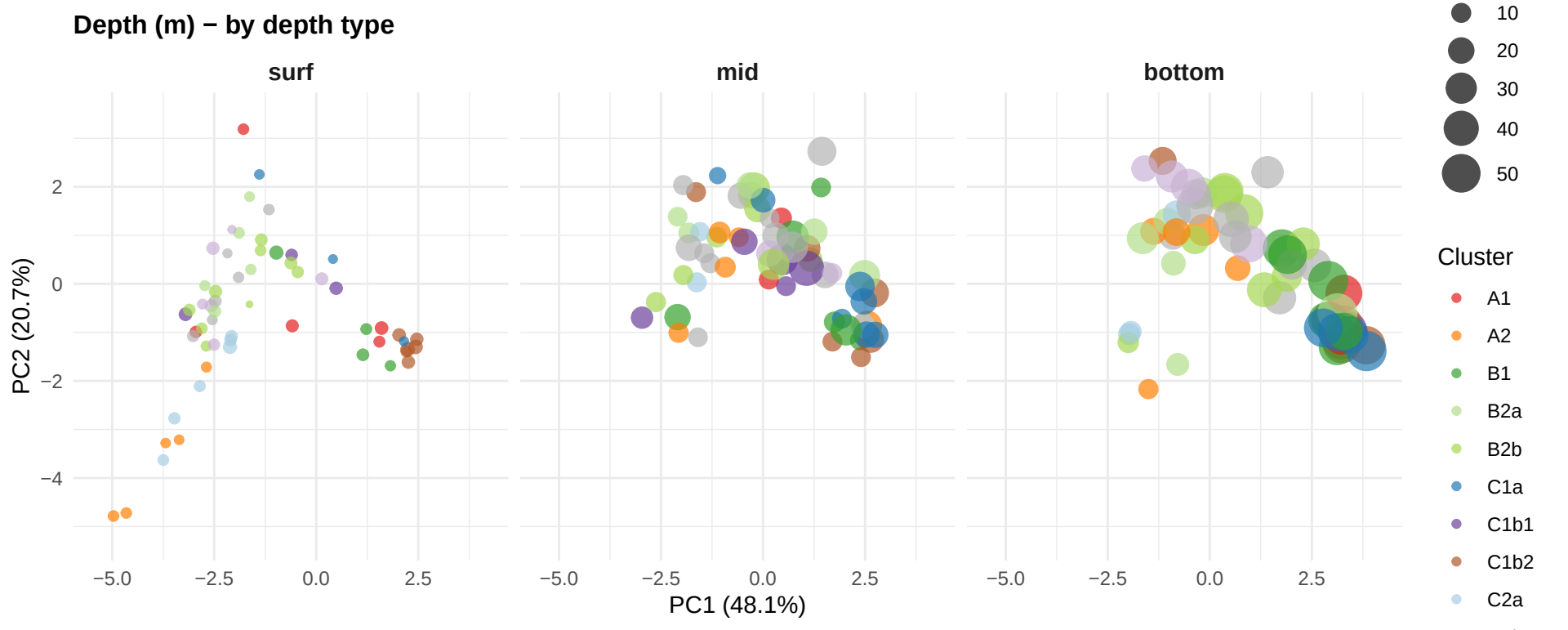

NH4 (µM) − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
NH4 (µM)
0
1
2
3
4


NH4 (µM) − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
NH4 (µM)
0
1
2
3


FIECO−AFL (mg/m³) − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
5
10
15
20
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1


FIECO−AFL (mg/m³) − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
FIECO−AFL (mg/m³)
0
5
10
15
20
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a


Depth (m) − all depths
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
20
30
40
50
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1


Depth (m) − by depth type
surf
mid
bottom
PC2 (20.7%)
PC1 (48.1%)
2
0
−2
−4
−5.0
−2.5
0.0
2.5
Depth (m)
10
20
30
40
50
Cluster
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a